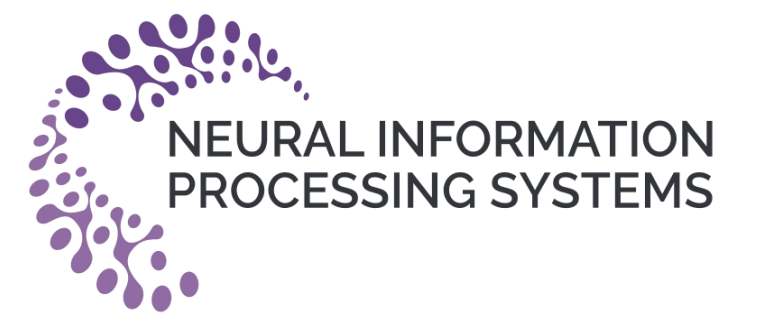


RollVerify: Bridging Efficiency and Accuracy in Long-Tail Rollout Reinforcement Learning



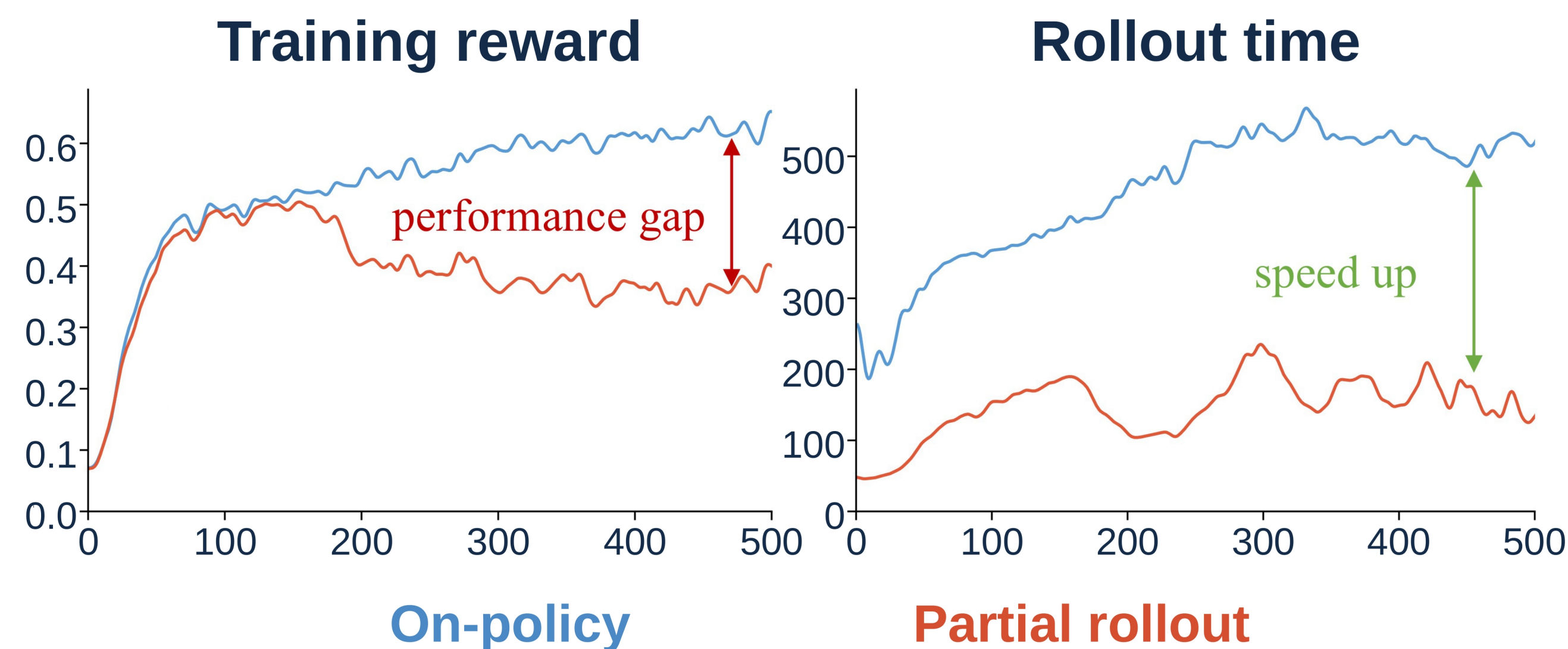
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01 Partial rollout trade-off

Less waiting, but cached samples become stale.

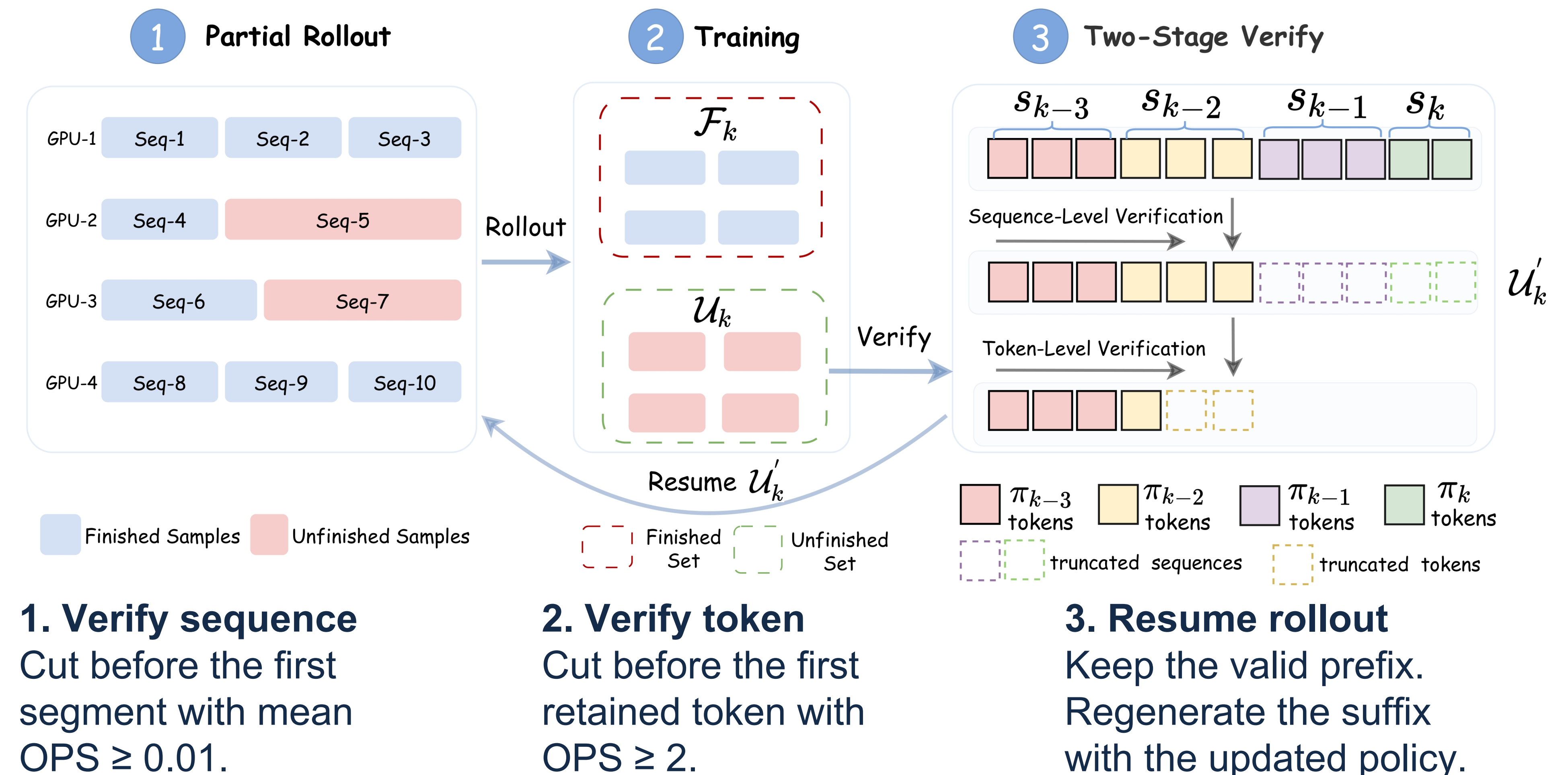


Off-Policy Shift (OPS)

$$D(y_t) = \left| \frac{\pi_\theta(y_t | x, y_{<t})}{\pi_{\text{old}}(y_t | x, y_{<t})} - 1 \right| \quad D(\tau) = \frac{1}{T} \sum_{t=1}^T D(y_t)$$

π_θ : current policy π_{old} : behavior policy T : trajectory length

02 Two-stage prefix verification



03 Accuracy at lower cost

1.7× faster with comparable accuracy

Model	Method	AVG $\uparrow$	GPU days $\downarrow$
Qwen3-8B-Base	On-policy	57.3	98.1
	RollVerify	57.2	57.5
Qwen3-30B-A3B-Base	On-policy	62.5	112.0
	RollVerify	62.5	65.1

Training cost (GPU days)

